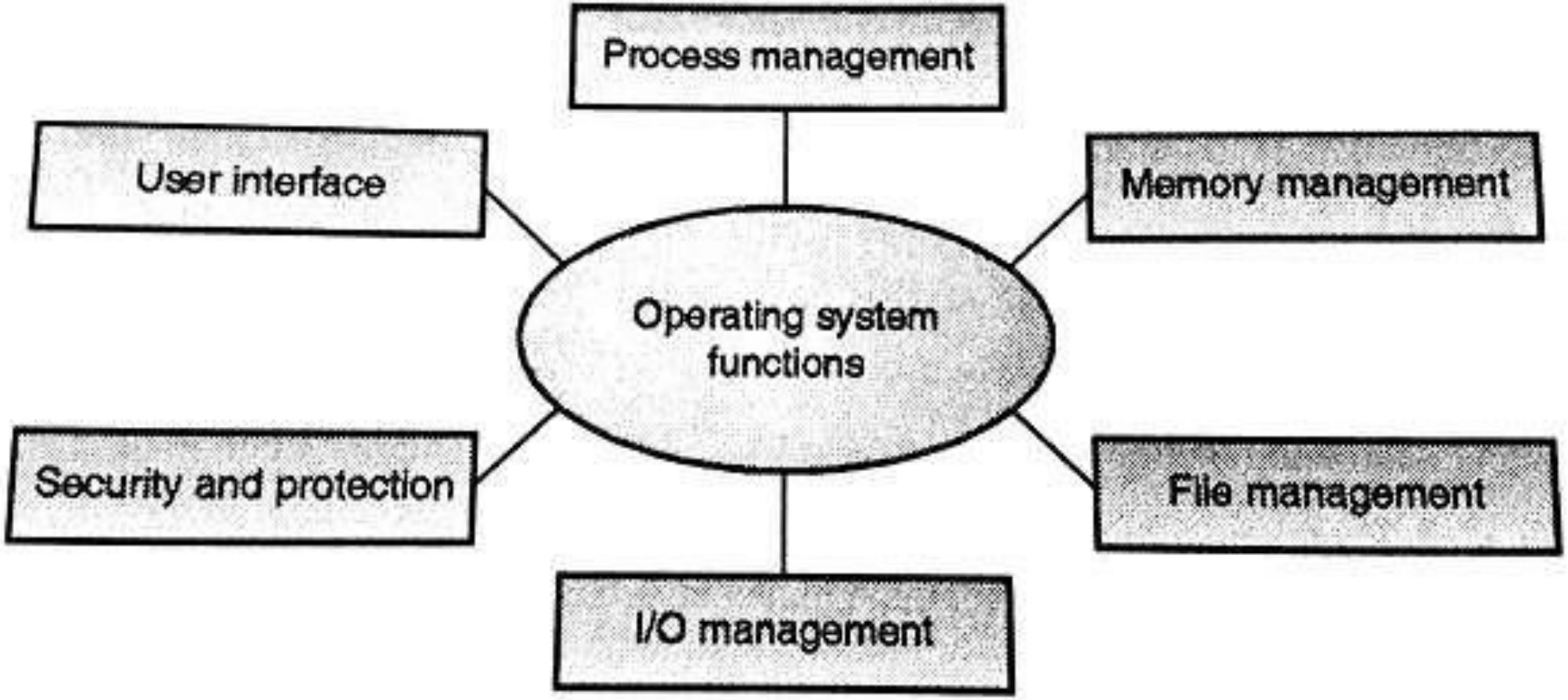
Ch. 6 Windows Operating Systems

IOCS – CS101

Functions of Operating System



Functions of Operating System

- **Memory Management:** The operating system manages the computer's primary memory and provides mechanisms for optimizing memory usage.
- **Resource Management:** The operating system manages and allocates memory, CPU time, and other hardware resources among the various programs and processes running on the computer.

Functions of Operating System

- **Process Management:** The operating system is responsible for starting, stopping, and managing processes and programs. It also controls the scheduling of processes and allocates resources to them.
- **Security:** The operating system provides a secure environment for the user, applications, and data by implementing security policies and mechanisms such as access controls and encryption.

Functions of Operating System

- **Job Accounting:** It keeps track of time and resources used by various jobs or users.
- **File Management:** The operating system is responsible for organizing and managing the file system, including the creation, deletion, and manipulation of files and directories.

Functions of Operating System

- **Device Management:** The operating system manages input/output devices such as printers, keyboards, mice, and displays. It provides the necessary drivers and interfaces to enable communication between the devices and the computer.
- **Networking:** The operating system provides networking capabilities such as establishing and managing network connections, handling network protocols, and sharing resources such as printers and files over a network.

Functions of Operating System

- **User Interface**: The operating system provides a user interface that enables users to interact with the computer system. This can be a Graphical User Interface (GUI), a Command-Line Interface (CLI), or a combination of both.
- **Backup and Recovery:** The operating system provides mechanisms for backing up data and recovering it in case of system failures, errors, or disasters.

Functions of Operating System

- **Virtualization:** The operating system provides virtualization capabilities that allow multiple operating systems or applications to run on a single physical machine. This can enable efficient use of resources and flexibility in managing workloads.
- **Performance Monitoring:** The operating system provides tools for monitoring and optimizing system performance, including identifying bottlenecks, optimizing resource usage, and analyzing system logs and metrics.

Functions of Operating System

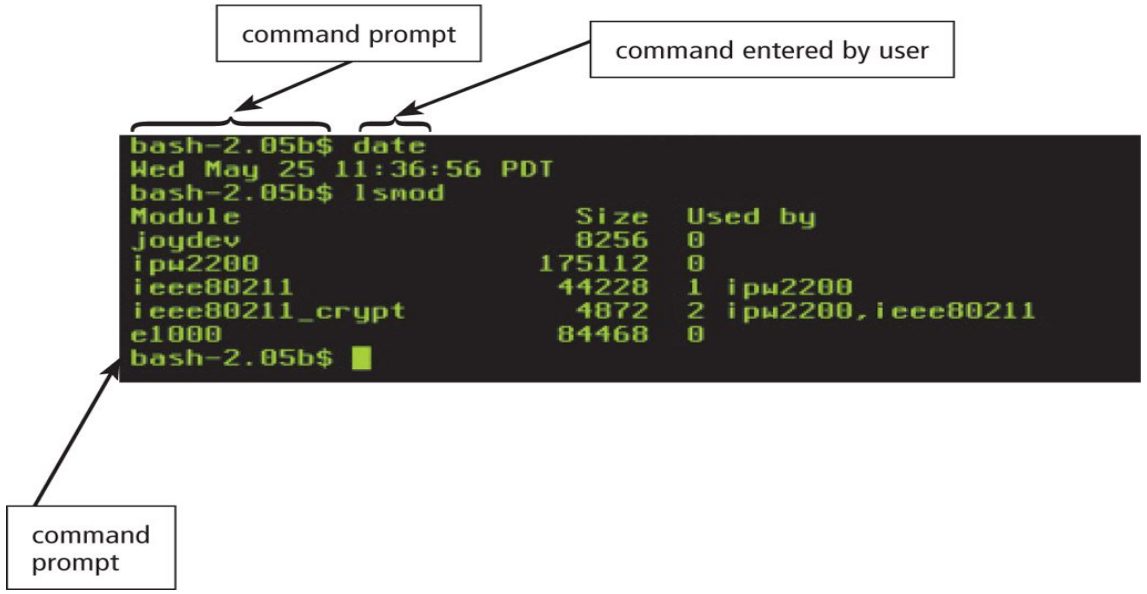
- **Time-Sharing:** The operating system enables multiple users to share a computer system and its resources simultaneously by providing time-sharing mechanisms that allocate resources fairly and efficiently.
- **Error-detecting Aids:** These contain methods that include the production of dumps, traces, error messages, and other debugging and error-detecting methods.

User Interface Mechanism

- A **user interface** (**UI**) controls how you enter data and instructions and how information is displayed on the screen
- There are two types of user interfaces
  1. Command Line Interface
  2. Graphical user Interface

Command-line interface .1

- In a command-line interface, a user types commands represented by short keywords or abbreviations or presses special keys on the keyboard to enter data and instructions



Graphical User Interface .2

- With a graphical user interface (GUI), you interact with menus and visual images



Microsoft Windows

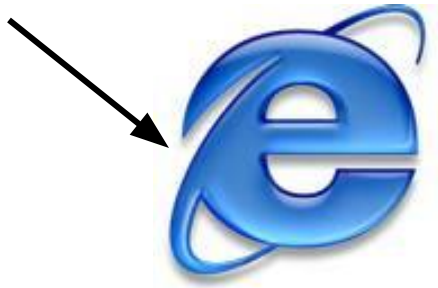
- The graphical Microsoft operating system designed for Intel-platform desktop and notebook computers.
- Best known, greatest selection of applications available.
- Current editions include Windows 7(ended on January 14, 2020)

, 8, 8.1,10 and 11.



?What Does Windows Do

- Provides an easy way to start programs (*the Start Button does this*)
- Runs more than one program at a time (*called Multitasking*)
- Provides an easy way to switch between open programs (*the taskbar does this*)
- Provides access to the Internet (*Internet Explorer does this*)



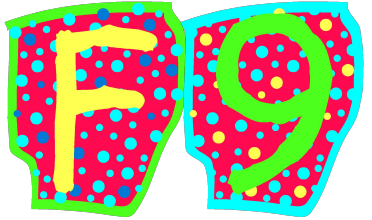
Common Mouse Pointer Shapes & What They Do

	Normal Select		Busy		Resizing
	Help Select		Move		Text Select
	Working in Background		Unavailable		Link Select





Controlling Windows via the keyboard



- **Keyboard** – used to enter data and to issue commands to the computer.
- Keys on the Keyboard:
  - **Function Keys**: the twelve keys located across the top of the keyboard that are labeled F1-F12. They perform specific tasks depending on the software you are using.
  - **Control Keys**: are Shift, Ctrl, and Alt. They are used in *combination* with other keys to issue commands.
  - **Keyboard Shortcuts**: are key combinations (usually using Control Keys) that can be used to activate certain commands as an alternative to using the mouse.

CommonKeyboard Shortcuts

- New ……Ctrl + N
- Open ……Ctrl + O
- Save ……..Ctrl + S
- Print ……..Ctrl + P
- Cut ……Ctrl + X
- Copy ……Ctrl + C
- Paste ……Ctrl + V
- Select All ……..Ctrl + A



Configuration of GUI Based OS

- **Button:** A graphical representation of a button that acts as a program when pressed.
- **Dialogue box:** A type of window that displays additional information and asks a user for input.
- **Icon:** Small graphical representation of a program, features, or file.
- **Menu:** List of commands or choices offered to the user through the menu bar.
- **Ribbon:** Replacement for the file menu and toolbar that groups programs activities together.
- **Tab:** Clickable area at the top of a window that shows another page or area.
- **Toolbar:** Row of buttons, often near the top of an application window, that controls software functions.
- **Window:** Rectangular section of the computer's display that shows the program currently being used.
- **Display server:** Coordinates the input and output between the operating system, hardware, and clients
- **Windowing system:** Provides the framework for a GUI environment, such as moving and drawing windows

Configuration of GUI Based Windows OS

- Open the Control Panel by clicking Start
- Select Network and Sharing Center or View network status and tasks
- Select Manage network connections
- Right-click on the Local Area Connection for your network controller
- Select Properties
- Click OK twice, then Close
- Repeat the steps for your wireless network connection if available
- Restart your computer for the changes to take effect

Configuration of Ethernet Settings

- 1. Connect Ethernet cable.
- 2. Go to Control Panel > Network and Sharing Center.
- 3. Click Change adapter settings.
- 4. Right-click Ethernet adapter > Properties.
- 5. Select Internet Protocol Version 4 (TCP/IPv4).
- 6. Enter IP address, subnet mask, gateway, and DNS servers.
- 7. Click OK.

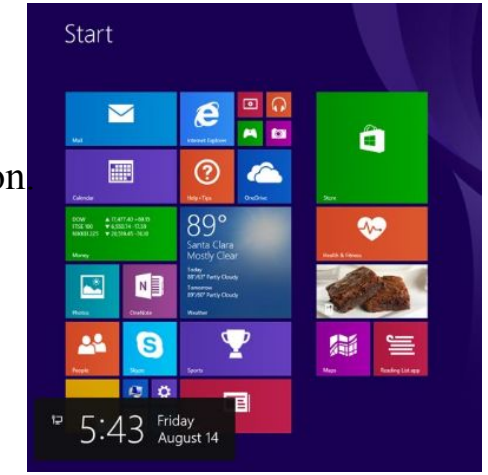
Windows Desktop, Tools and Applications

- Windows Desktop
  - A graphical representation of the workspace, often called GUI.
  - The desktop has icons, toolbars, and menus to manipulate files.
  - Windows 8 introduced a new desktop that uses tiles.
- Desktop Properties
  - The Windows desktop is highly customizable.
  - Possible customizations include: re-arrange titles, change colors and sounds, modify screen resolution.
- Start Menu
  - Allow users to manage programs, search the computer, and manipulate running applications.
  - The Start Menu can accessed by clicking the Windows icon.
  - Microsoft removed the Start Menu from Windows 8 but added back a limited version of it in Windows 8.1.



Windows Desktop, Tools and Applications (Cont.)

- Task Manager
  - Allows for the management of all applications running on the computer.
  - While presenting essentially the same information, Windows 8 has different tabs than Windows 7 & Vista.
- Computer and Windows Explorer
  - File Explorer and Windows Explorer are similar file manager applications.
  - This PC and Computer features grant access to various drives.
- Windows Libraries
  - Allows for easy organization of content; no need to move files.
  - Presents content from different locations in the same folder.
- Install and Uninstall Applications
  - The installation process copies all necessary files to the proper disk location.
  - Setup.exe usually starts Windows installation processes.
  - The uninstall process removes the application and its related files.



Control Panel Utilities

- Introduction to Control Panel Utilities
  - Centralizes the settings for many features.
  - Settings are categorized in utilities, or small programs.
- User Accounts
  - Allows for account management.
  - UAC monitors programs and issues alerts upon system threating actions.
- Internet Options
  - Allows for configuring Windows Internet settings.
  - Internet Options tabs are General, Security, Privacy, Content, Connections, Programs, Advanced.
- Display Settings
  - Allows for configuring video-related settings.
  - Screen resolution, orientation, single/multiple displays are a few of the settings available under Display Settings.



Control Panel Utilities (Cont.)

- Folder Options
  - Allows for configuring folder-related settings.
  - Allows for adjusting the behavior of folder browsing, how files are opened and displayed and more.
- Action Center
  - Centralized location to view alerts that help keep Windows running smoothly.
  - Divided into Security and Maintenance sections.
  - Messages highlighted in red should addressed quickly; yellow-highlighted items indicate recommended tasks.
- Windows Firewall
  - Helps prevent malicious attacks to the computer.
  - Allows for adjusting firewall settings for home, work or public networks.
  - Firewall settings include allowing a specific program through the firewall, turning the firewall on or off and managing firewall notifications.



Control Panel Utilities (Cont.)

- Power Options
  - Allows for adjusting power consumption.
  - Power plans can be customized to control how the computer and its components use power.
- System Utility
  - Acts as a system information hub.
  - Allows users to view basic system information, access tools, and configure advanced system settings.
- Device Manager, Devices and Printers and Sound
  - Device Manager provides a list of all the devices installed in the computer and their statuses.
  - Device Manager is a useful troubleshooting tool.
  - Devices and Printers displays devices connected to the computer through a port or network connection.
  - The Sound utility allows for configuring audio devices or changing the sound scheme of the computer.



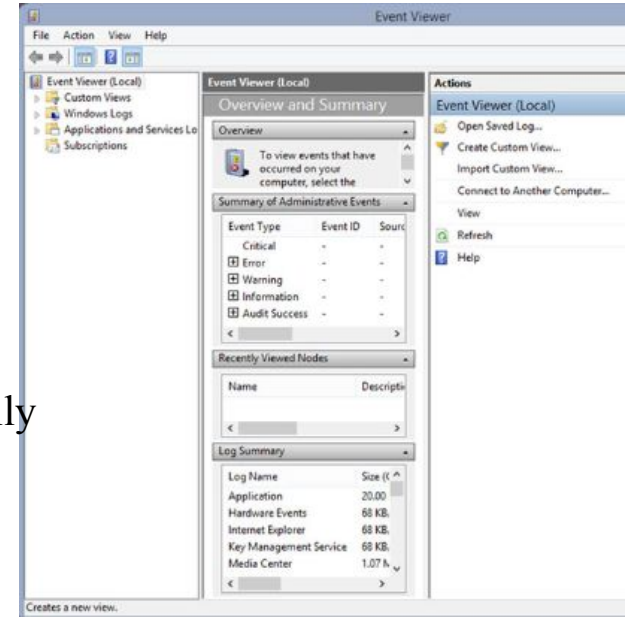
Control Panel Utilities (Cont.)

- Region, Programs and Features and Troubleshooting
  - Region (Win8) allows for changing the format of numbers, currencies, dates, and time.
  - Language (Win8) allows for changing the primary language or install an additional language.
  - In Win 7 & Vista both tools are combined into the Region and Language tool.
  - Programs and Features allows for uninstalling programs or for changing a program's configuration.
  - The Troubleshooting tool has a number of built-in scripts to identify and solve problems.
- HomeGroup and Network and Sharing Center
  - HomeGroup allows Windows computers to share resources.
  - HomeGroup is available on Windows 7 or higher.
  - Network and Sharing Center allows for configuring and reviewing all network operations in a Windows computer.



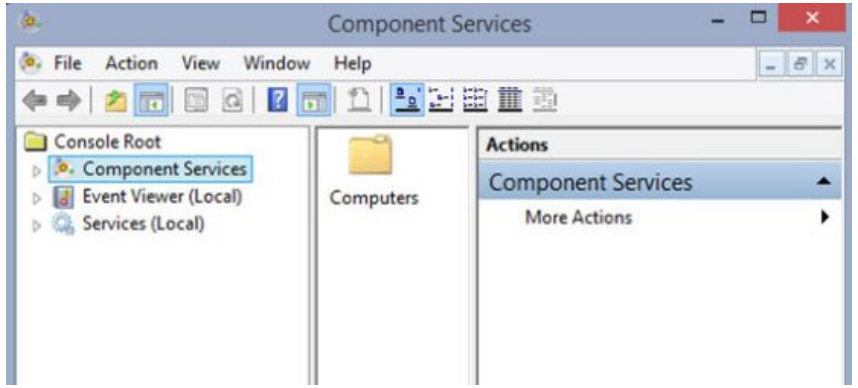
Administrative Tools

- Computer Management
  - Allows for the management of many aspects of the computer and remote computers in one tool.
  - Provides access to a number of utilities: Task Scheduler, Event Viewer, Shared Folders, Local Users and Groups, Performance, Device Manager and Disk Management.
- Event Viewer
  - Logs the history of application, security, system events.
  - These log files are a crucial when troubleshooting.
- Services
  - Allows for the management of all the services on your computer and remote computers.
  - Services can be configured to start automatically, automatically with delay, or manually.



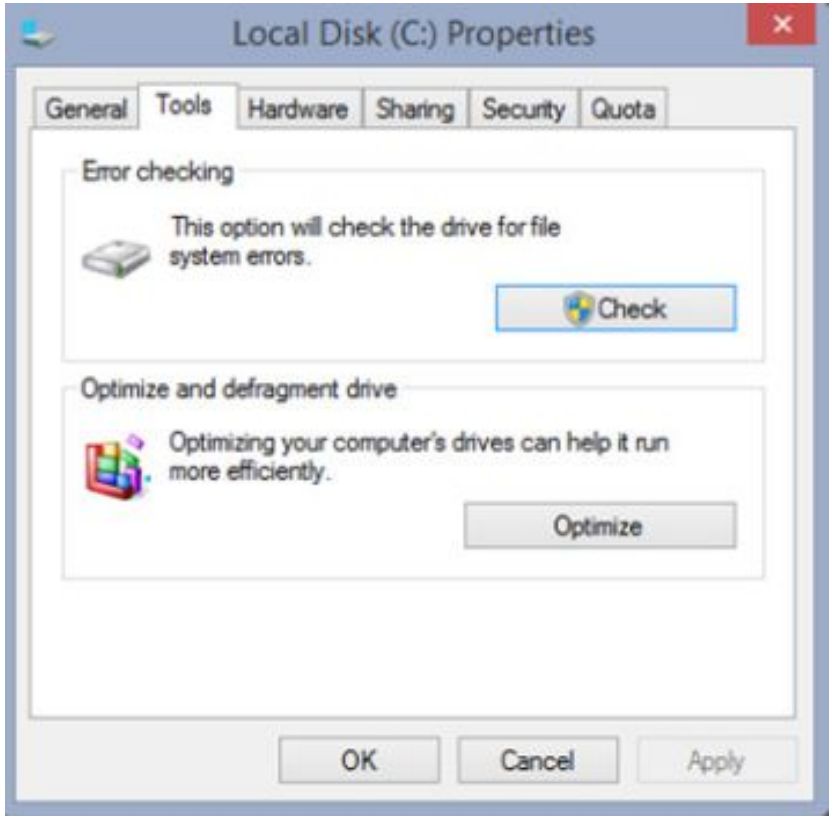
Administrative Tools (Cont.)

- System Configuration
  - Used to identify problems that keep Windows from starting correctly.
  - Services and startup programs can be turned off to help troubleshooting.
- Performance Monitor and Windows Memory Diagnostics
  - Displays an overview of the Performance Monitor and System Summary.
  - System Summary displays real-time information about the processors, disks, memory, and network usage.
  - System Summary displays detailed data about the resources currently in use.



Disk Defragmenter and Disk Error-Checking Tool

- Disk Defragmenter and Disk Error-Checking Tool
  - Disk Defragmenter consolidates files for faster access.
  - SSDs do not need user-triggered defragmentation; the drive controller defrags as it writes to the disk.
  - Disk Error-Checking scans the hard drive for file structure or disk surface errors.
- System Information
  - Combines information about software, drivers, hardware configurations, and computer components in one place.
  - Helpful for support personnel when troubleshooting and diagnosing a computer.



Command Line Tools

- Windows Command Line Interface
  - CLI is a text-based interface.
  - CLI can be useful when troubleshooting OS problems, especially if the GUI is inoperative.
- System Utilities
  - Run Line utility allows for invoking a tool by typing its name; no need to find the icon and click it.
  - A few common tools include: COMMAND, DXDIAG, EXPLORER, MMC, MSINFO32, MSTSC, NOTEPAD, REGEDIT.

